



# Data Exploration: Descriptive Statistics

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# Basics of Sampling

Population versus sample:

- A *population* is the entire set of objects or events under study. Population can be hypothetical “all students” or all students in this class.
- A *sample* is a “representative” subset of the objects or events under study. Needed because it's impossible or intractable to obtain or compute with population data.



# Basics of Sampling

Biases in samples:

- ***Selection bias***: some subjects or records are more likely to be selected
- Volunteer/ ***nonresponse bias***: subjects or records who are not easily available are not represented

Examples?

# Sample Mean

The **mean** of a set of  $n$  observations of a variable is denoted  $\bar{x}$  and is defined as:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean describes what a “typical” sample value looks like, or where is the “center” of the distribution of the data.

Key theme: there is always uncertainty involved when calculating a sample mean to estimate a population mean.

# Sample Median

The *median* of a set of  $n$  number of observations in a sample, ordered by value, of a variable is defined by

$$\text{Median} = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{x_{n/2} + x_{(n+1)/2}}{2} & \text{if } n \text{ is even} \end{cases}$$

Example (already in order):

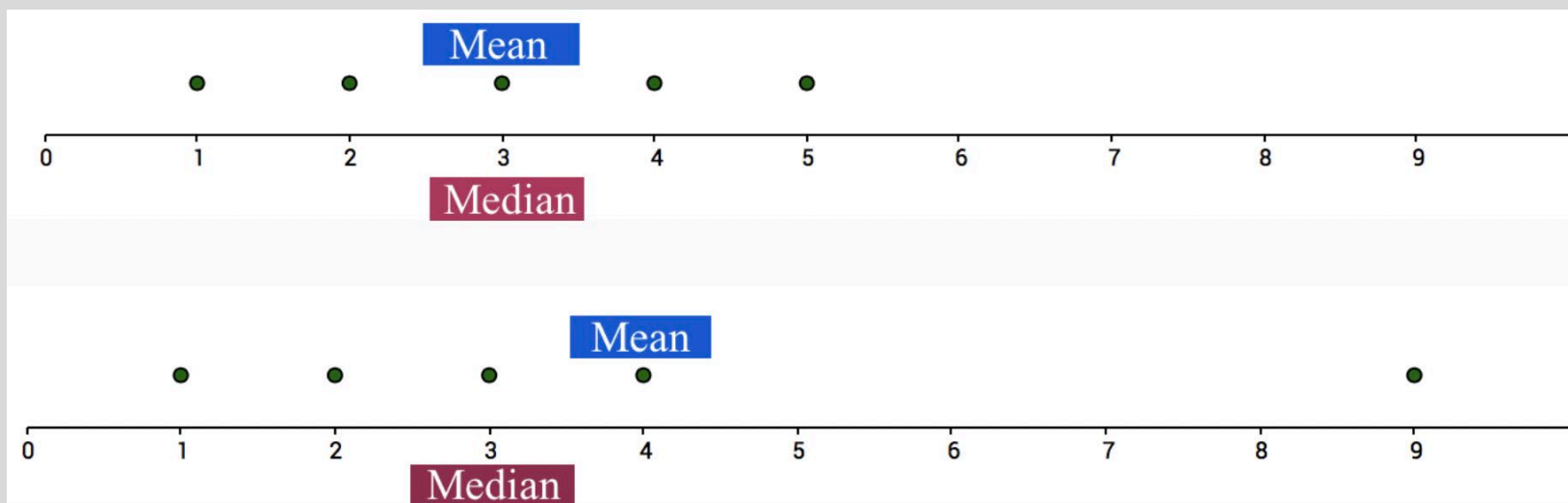
- Ages: 17, 19, 21, 22, 23, 23, 23, 38
- Median =  $(22+23)/2 = 22.5$

The median also describes what a typical observation looks like, or where is the center of the distribution of the sample of observations.



# Mean vs. Median

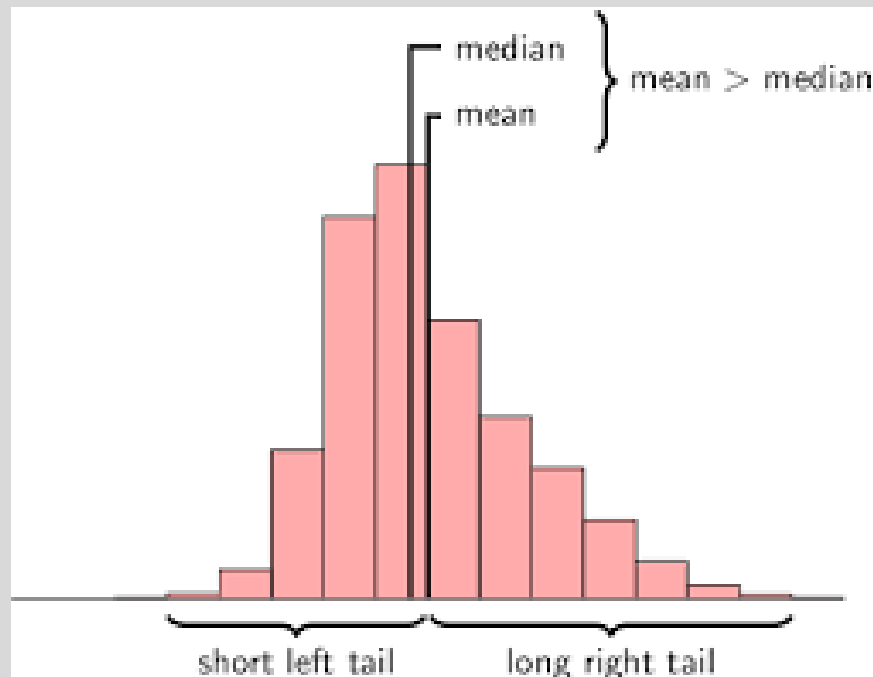
The mean is sensitive to extreme values (**outliers**)



# Mean, Median, and Skewness

The mean is sensitive to outliers:

The distribution is called **right-skewed** since the mean is greater than the median. Note: **skewness** often “follows the longer tail”.





# Computational Time

How hard (in terms of algorithmic complexity) is it to calculate

- the mean?
  - at most  $O(n)$
- the median?
  - at most  $O(n)$  or  $O(n \log n)$

Note: Practicality of implementation should be considered!

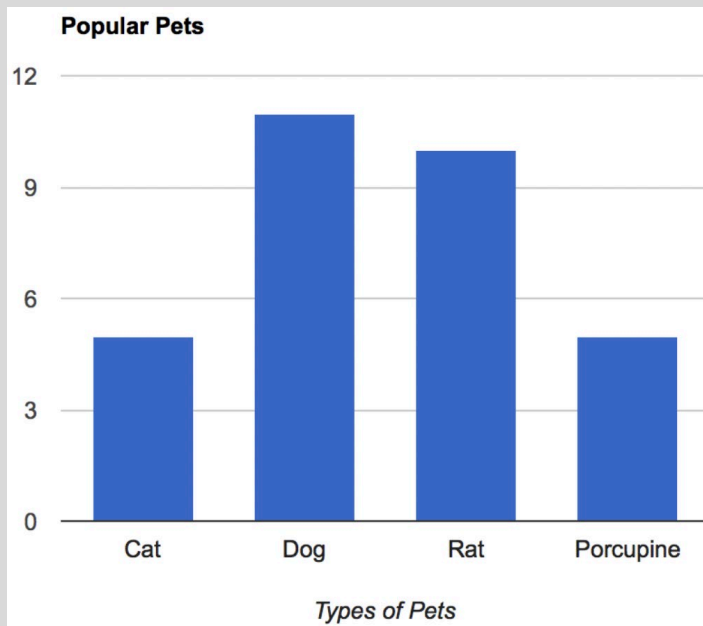




# Regarding Categorical Variables

For categorical variables, neither mean or median make sense. Why?

The mode might be a better way to find the most “representative” value.





# Measures of Spread: Range

The spread of a sample of observations measures how well the mean or median describes the sample.

One way to measure spread of a sample of observations is via the *range*.

Range = Maximum Value - Minimum Value

# Measures of Spread: Variance

The (sample) *variance*, denoted  $s^2$ , measures how much on average the sample values deviate from the mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n |x_i - \bar{x}|^2$$

Note: the term  $|x_i - \bar{x}|$  measures the amount by which each  $x_i$  deviates from the mean  $\bar{x}$ . Squaring these deviations means that  $s^2$  is sensitive to extreme values (outliers).

Note:  $s^2$  doesn't have the same units as the  $x_i$  :(  
What does a variance of 1,008 mean? Or 0.0001?



# Measures of Spread: Standard Deviation

The (sample) *standard deviation*, denoted  $s$ , is the square root of the variance

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n |x_i - \bar{x}|^2}$$

Note:  $s$  does have the same units as the  $x_i$ !



# Questions?